OCC: SOFTWARE ENGINEER YEAR 2

SAQA ID
NQF 6
Software design and testing C#
SDT621
WEEK 2 GROUP PROJECT
2026

CTU training solutions | Private Tertiary Education

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DECLARATION OF AUTHENTICITY

I _____ (GROUP MEMBER NUMNER)

hereby declare that the contents of this assessment _____

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: _____

Date: _____

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QUALIFICATION AND PROJECT DETAILS

Programme Title	OCC: SOFTWARE ENGINEER
Module Title	Software design and testing C#
Module Code	SDT621
NQF Level	6
Assessment Type	WEEK 2 GROUP PROJECT-1
Hand Out Date & Time	@10h30am - 22 / 0 4 / 2026
Hand In & Time	@11h59AM - 22 Avril 2026
Project Lead Developer	Lusukama Selemani
Submission code via	Github and share the link

Student Name and Surname	Group Member Number

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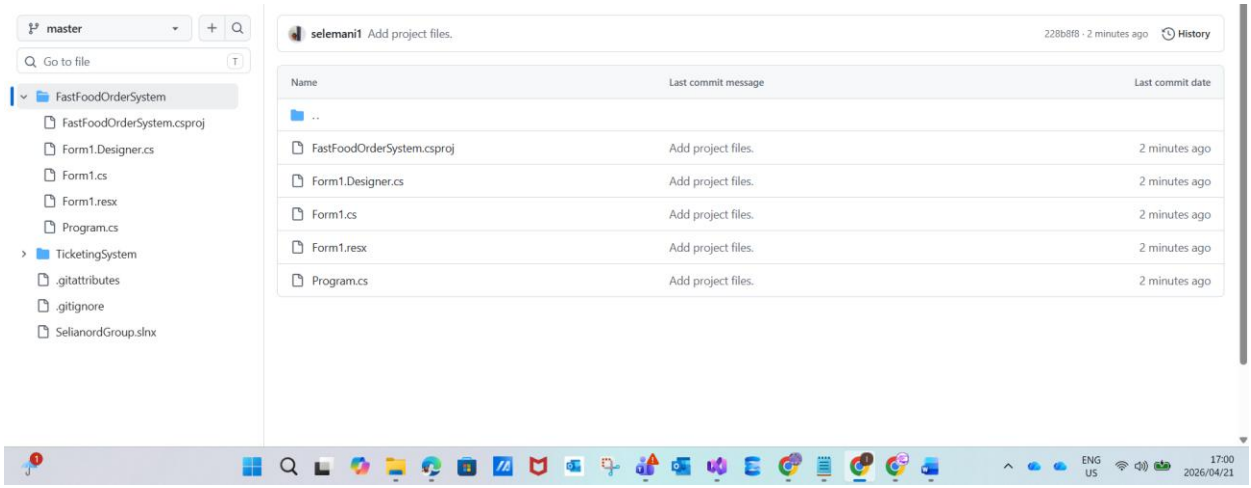
Example of Collecting Evidence for your submission pdf:

Question1 :



Ensure each screenshot shows the **full screen**, including the **system date and time**

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Group Project: Fast Food Ordering System with GitHub Collaboration

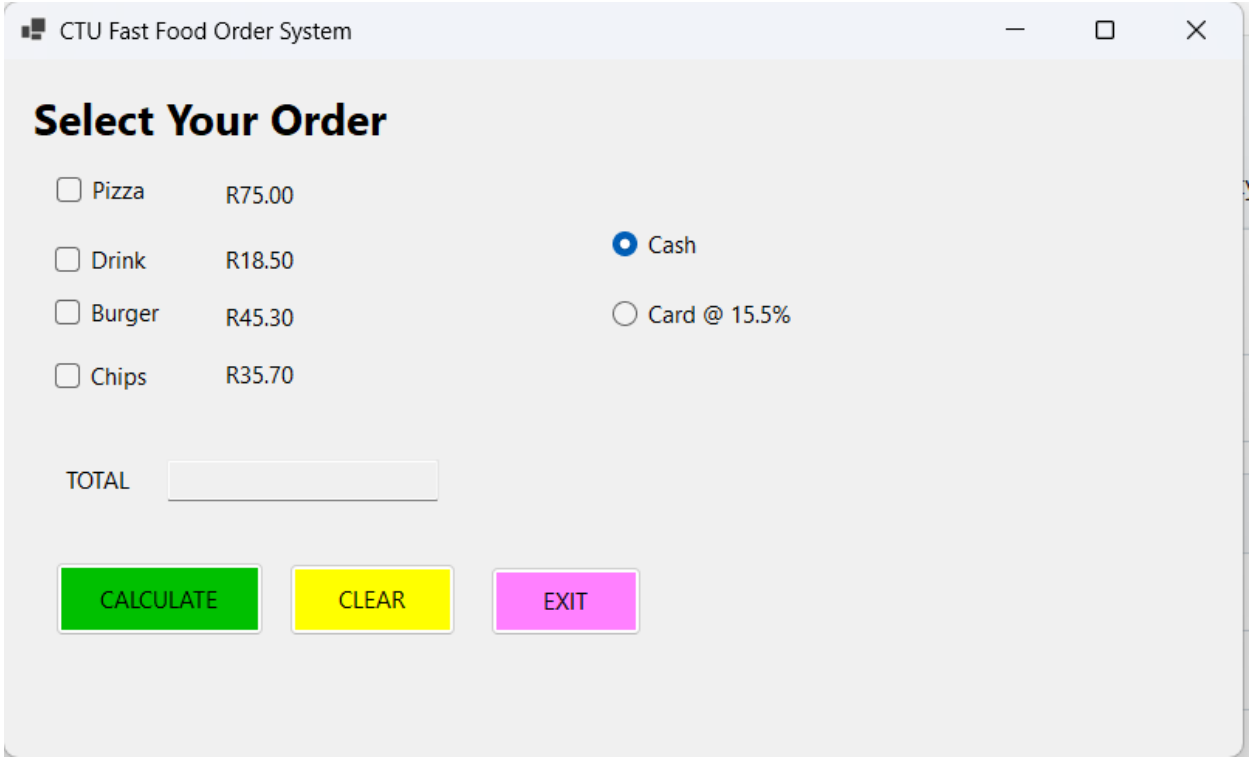
Fast Food Ordering System – Windows Forms App in C#

Group Size

- Minimum: 4 members
- Maximum: 5 members

Project Overview

You are required to work as a team to design and develop a Windows Forms Application in C# for a Fast Food Ordering System.



The system must allow a user to:

- select food items
- choose a payment method
- calculate the total amount
- clear the order
- exit the application

Your group must also use GitHub to manage teamwork, version control, and code integration.

Main System Requirements

The application must include the following:

Functional Requirements

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1. Display food items with prices:
    - Pizza
    - Drink
    - Burger
    - Chips
2. Allow the user to select items using **CheckBox** controls.
3. Allow the user to choose payment method using **RadioButton** controls:
    - Cash
    - Card @ 15.5%
4. Display the total amount in a textbox.
5. Include buttons for:
    - **Calculate**
    - **Clear**
    - **Exit**
6. If **Card** is selected, add **15.5%** to the total.
7. If **Clear** is clicked:
    - all food items must be unchecked
    - payment selection must reset
    - total textbox must clear

Minimum Technical Requirements

The group must demonstrate:

- C# Windows Forms development
- event-driven programming
- input processing
- control handling
- teamwork using GitHub
- version control using commits and branches

Group Member Roles and Tasks

Member 1: UI/UX Designer

Responsibilities:

- create the Windows Form layout
- add labels, checkboxes, radio buttons, buttons, and textbox
- align controls neatly
- improve the visual appearance of the application

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- ensure the form is user-friendly

Deliverables:

- completed interface design
- properly named controls
- screenshot of designed form

Member 2: Business Logic Developer

Responsibilities:

- write the calculation logic
- assign correct prices to menu items
- implement card payment charge of 15.5%
- ensure total is displayed correctly
- test calculation scenarios

Deliverables:

- working calculation code
- tested total results
- code comments explaining the logic

Member 3: Validation and Reset Developer

Responsibilities:

- implement clear/reset logic
- ensure total textbox clears properly
- ensure payment method resets correctly
- validate system behavior for empty selections
- help handle application exit functionality

Deliverables:

- working Clear and Exit features
- clean reset behavior
- test evidence for reset functionality

Member 4: GitHub and Documentation Manager

Responsibilities:

- create the GitHub repository
- add all group members as collaborators
- manage README file
- ensure members use commits properly
- document project setup and usage
- collect screenshots of commit history

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- prepare final submission evidence

Deliverables:

- GitHub repository link
- README.md file
- commit history evidence
- contribution summary for each member

Member 5: Tester / Quality Assurance

Responsibilities:

- test all features of the system
- identify bugs and issues
- verify calculations
- verify UI usability
- produce a short testing report

Deliverables:

- test cases
- bug report
- retest evidence after fixes

GitHub Requirements

Each group must use **GitHub** properly.

GitHub Rules

1. One member creates the repository.
2. All group members must be added as collaborators.
3. Each member must make their **own contribution**.
4. Each member must make **at least 3 meaningful commits**.
5. Commit messages must be clear, for example:
    - Created initial form layout
    - Added calculation logic
    - Fixed clear button issue
    - Updated README documentation
6. Use branches where possible, for example:
    - Dev-ui-design-YourName
    - Dev-calculation-logic-YourName
    - Dev-validation-feature-YourName
    - Dev-documentation-YourName
7. Final code must be merged into the **main branch**.

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